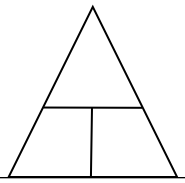
Fill in the triangle



Power Calculations Worksheet

power = $\frac{\text{work done}}{\text{time}}$

$P = \frac{W}{t}$

Units: Power in W (_____)
Work in J (_____)
time in s (_____)

1. Find the unknown quantity:

a) W = 10 J P = 100 W t = ?	b) W = ? P = 75 W t = 20 s	c) W = 15 kJ P = ? t = 20 s

2. Find the unknown quantity (convert to the correct units first)

a) W = 1.2 kJ = _____ J P = 75 W t = ?	b) W = 55 J P = ? t = 2.25 hrs = _____ s	c) W ? P = 200 MW = _____W t = 0.5 hrs = _____s

WORD PROBLEMS

1. How much work does your bedroom lamp (containing a 100 W light-bulb) do every evening when you read for half an hour before you go to bed at 9:45pm?

2. How much work is done pulling a fully laden sled with a force of 150N for 15 seconds over 3m?

3. How long will a 75 W bulb glow for on 1 kJ of energy?